

Executive Summary: Used-Car Pricing & Market Valuation Analysis

Project Title: Car Price Analysis — SQL Portfolio Project

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Database System: MySQL 8.0

Dataset Scope: 6,643 Indian Used Vehicle Listings (1983–2020)

1. Project Background & Business Context

The secondary automotive market presents complex pricing dynamics influenced by vehicle age, accumulated mileage, powertrain options, and ownership history. This project analyzed **6,643 used car listings** to evaluate key price drivers, identify depreciation patterns, and discover value opportunities.

By employing **MySQL 8.0** for regex-based text parsing, schema modification, conditional aggregations, and CTE-driven ratio analyses, this study provides data-backed recommendations for inventory managers, pricing analysts, and individual sellers.

2. Core Analytical Findings

Dimension / Variable	Analytical Finding	Business Implication
Transmission Type	Automatic vehicles average ₹1,322,984 vs. ₹455,306 for manual vehicles (~3x premium).	Automatics represent luxury/upper-segment inventory and warrant dedicated financing packages.
Seller Channel	Dealer listings average ₹935,000 , while private seller listings average ₹484,000 (~2x gap).	Dealer markups reflect reconditioning and trust; private sellers lose equity without third-party validation.

Powertrain / Fuel	Diesel models average ₹648,567 vs. ₹387,977 for Petrol models (+67% premium).	Diesel engines maintain capital value far longer over time due to buyer demand for efficiency.
Ownership Tier	Each additional owner reduces resale valuation by ₹100,000 – ₹200,000 .	Depreciation steepens past 1st ownership; multi-owner cars require faster inventory turn times.

3. Key Strategic Recommendations

High Priority

1. **Targeted Financing for Automatics:** Develop specialized financing and extended warranty bundles for automatic inventory to capture buyer willingness to pay in the ₹1.3M+ pricing tier.
2. **Third-Party Certification for Private Sellers:** Offer a paid "Vehicle Condition & Inspection Certificate" product for individual sellers to help bridge the ₹450K price gap against dealer listings.

Medium Priority

3. **Optimized Procurement for Used Diesel Inventory:** Prioritize second-hand diesel vehicles with verified service histories, pricing them confidently above petrol equivalents even as vehicle age increases.

Low Priority

4. **Dynamic Trade-In Valuation Rules:** Adjust trade-in pricing algorithms to apply non-linear valuation markdowns as soon as a vehicle transitions to second- or third-owner status.

4. Technical Architecture & Data Pipeline



